Master Thesis

The Impact of Portfolio Construction Methods on the Shape of Long-Run Return Distributions

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Abstract

The choice of portfolio construction method can have a statistically significant effect on the skewness of long-run return distributions. For small portfolios, the benefit of diversifying more outweighs the benefit of changing the way a portfolio is constructed. For larger portfolios it can be worth changing from an equal-weighted approach to a minimum variance or risk parity approach. These findings show that investors should make sure they are not underdiversified before worrying about how they construct their portfolio, and they illustrate that the level of skewness an investor is exposed to can be influenced by their investment decisions.

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Auftragserteilung für die Masterarbeit

Sehr geehrter Herr Öztürk

Mit Freude habe ich erfahren, dass Sie bei mir die Masterarbeit schreiben möchten. Hiermit erteile ich Ihnen dafür den folgenden Auftrag:

Topic: The Impact of Portfolio Construction Methods on the Shape of Long-Run Return Distributions

Die geplante Masterarbeit untersucht, wie verschiedene Methoden der Portfoliokonstruktion die langfristige Verteilung von Aktienrenditen beeinflussen. In der Literatur ist gut dokumentiert, dass Aktienrenditen über lange Horizonte stark asymmetrisch verteilt sind: Nur wenige Gewinneraktien erklären den Großteil der Wertschöpfung, während die Mehrheit der Titel den Markt kaum oder gar nicht übertrifft. Diese Schiefe (Skewness) der Renditeverteilung stellt Investoren vor besondere Herausforderungen, da der Mittelwert als klassisches Bewertungsmaß das tatsächliche Risiko- und Chancenprofil oft nur unzureichend widerspiegelt.

Vor diesem Hintergrund soll die Arbeit analysieren, inwiefern die Wahl der Portfoliokonstruktionsmethode (z. B. Equal Weight, Value Weight, Minimum Variance, Risk Parity) systematisch Einfluss auf die Form der langfristigen Renditeverteilungen hat. Methodisch kann dies entweder durch eine direkte Untersuchung der Skewness oder über ein erweitertes Nutzenmass erfolgen.

Die Arbeit beginnt mit einer klaren und prägnanten Einleitung. Diese sollte den Hintergrund des Themas, die Forschungsfragen, die Methodik sowie den erwarteten Beitrag der Arbeit darstellen. Besonders wichtig ist es, die Aktualität und Relevanz des Themas hervorzuheben. Diskutieren sie in der Einleitung auch kurz ihre wichtigsten Erkenntnisse.

Im zweiten Teil erfolgt eine systematische Aufarbeitung der relevanten wissenschaftlichen Literatur. Der Schwerpunkt liegt auf hochwertigen akademischen Publikationen. Die Literaturanalyse soll nicht nur bestehende Studien zusammenfassen, sondern diese kritisch einordnen und Verbindungen zu den eigenen Forschungsfragen herstellen. Bitte verwende das Zitierformat des Journal of Finance und orientiere dich im Schreibstil an publizierten wissenschaftlichen Arbeiten.

Im Anschluss werden die verwendeten Daten für die empirische Analyse vorgestellt. Die Auswahl der Daten ist nachvollziehbar zu dokumentieren, inklusive einer genauen Beschreibung der Datenquellen. Ziel ist ein hohes Mass an Transparenz, um die Reproduzierbarkeit der Ergebnisse zu gewährleisten.

Der Hauptteil der Arbeit widmet sich der empirischen Analyse der verschiedenen Ansätze zur Portfoliokonstruktion und ihr Einfluss auf die Renditeverteilung oder den Investorennutzen. Beschreiben und begründen Sie ihr Vorgehen und die verwendeten Methoden.

Bei der Ableitung der Ergebnisse ist das methodische Vorgehen gemäß den Absprachen mit dem Betreuer sorgfältig zu dokumentieren. Programmierarbeiten und statistische Befehle (z. B. aus Python, R, Stata, Matlab oder SAS) sind verständlich zu erläutern. Sämtliche Quellen müssen sowohl in gedruckter als auch in elektronischer Form verfügbar sein, um eine Überprüfung und Replizierbarkeit der Resultate sicherzustellen.

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Bitte beachten Sie folgende formale Kriterien:

- Sie verfassen die Arbeit in englischer Sprache.
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    - Titelblatt
    - Abstract nach dem Vorbild wissenschaftlicher Zeitschriften (max. 100 Worte)
    - Vorliegende Auftragserteilung
    - Inhaltsverzeichnis
    - Hauptteil
- Lade auf OLAT ein ZIP-File mit folgenden Dateien:
    - Titelblatt der Arbeit (PDF-Dokument)
    - Executive Summary (PDF-Dokument): Zusammenfassung auf max. 3 Seiten
    - Gesamte Arbeit (LaTex- oder Word-File sowie PDF-Dokument)
    - Gesamtes verwendetes Datenmaterial
    - Computer-Codes zur Replikation Ihrer Ergebnisse
    - Elektronisch gespeicherte Referenzen (Papers in PDF-Format).

Bei allfälligen Fragen wenden Sie sich an Sandro Braun, Institut für Finance,
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Gute Arbeit und viel Erfolg!

Freundliche Grüsse

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Selbstständigkeitserklärung

Hiermit erkläre ich, dass ich die vorliegende Arbeit selbständig und ohne Benutzung anderer als der angegebenen Hilfsmittel angefertigt habe. Mir ist bekannt, dass ich selbst die volle Verantwortung für die Wissenschaftlichkeit des vorgelegten Textes übernehme, auch wenn generative KI-Hilfsmittel wie ChatGPT oder andere künstliche Intelligenz eingesetzt wurden. Alle Stellen, die wörtlich oder sinngemäss aus veröffentlichten oder nicht veröffentlichten Schriften entnommen wurden, sind als solche kenntlich gemacht. Die Arbeit ist in gleicher oder ähnlicher Form oder auszugsweise im Rahmen einer anderen Prüfung noch nicht vorgelegt worden.

Kenan Öztürk

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1 Introduction

The tremendous success of Markowitz (1952), which laid the groundwork for modern portfolio theory, directed the main focus of the financial literature in the area of investment portfolio optimization on the mean and variance of returns. The prevailing paradigm for many decades has been to maximize the expected return for a given level of risk, or equivalently, to minimize the risk for a given expected return. This approach however ignores the higher moments of the return distribution completely, even tough we know since at least Arditti and Levy (1975), that multiperiod returns are positively skewed. So, ignoring the third standardized moment (skewness) can present a problem many investors conveniently ignore.

To see why skewness can be a problem, consider Bessembinder et al. (2023), who argue that the existence of strong positive skewness has important implications for financial planning. They give the example of pension funds who, when assessing whether they are adequately capitalized, make assumptions based on the mean of possible future returns. They argue that this is inappropriate if the distribution is positively skewed because then the majority of outcomes will be less than the mean. Similarly, Farago and Hjalmarsson (2023a) argue that for the purpose of saving for retirement, the relevant horizon can easily span several decades, and that understanding long-run skewness of total compound returns is therefore of great importance. Thus, I argue that understanding how and if the method by which a portfolio is constructed impacts long-run skewness is of great interest to both institutional and individual investors. In particular, they want to know if the portfolio construction method they consider using could have a negative effect on the distribution of long-run outcomes. And as I find, investors who follow an equal weighted approach do well already, as long as they are sufficiently diversified.

According to Farago and Hjalmarsson (2019), diversification is effective at reducing skewness because it reduces the standard deviation of short-run returns. Given that knowledge, I argue that a minimum variance approach might be even better suited to decrease skewness than a simple equal-weighted approach, because it reduces that standard deviation even further. In general, the choice of portfolio construction method undoubtedly impacts the volatility of short-run portfolio returns, and it is therefore conceivable that the resulting long-run skewness is also influenced. This is the core idea behind my main research question, which is: *Can the (optimization) method used to construct a stock portfolio have a significant effect on the skewness of long-run return outcomes?*

Furthermore, I am not only interested in how the skewness of returns is affected, but also in whether a potential reduction in skewness comes at a significant cost. A cost in this sense might be a sacrifice of the mean return. For this purpose, I define a ratio that takes into account the mean, the standard deviation and the skewness of long-run returns. I call this metric the return-to-shape (RTS) ratio, because it relates the mean return to the shape of the distribution, which I define as a combination of the standard deviation and skewness.

To conduct this analysis, I bootstrap portfolio returns from historical stock returns (US stock market, from 1998 to 2026) using a bootstrap procedure proposed and used by Bessembinder (2018). A distribution of outcomes is obtained by repeating the bootstrap procedure 2'000 times, and its mean, standard deviation and skewness are computed. This is done for multiple portfolio sizes, investment horizons and portfolio construction methods. Finally, I estimate two regression models (one for skewness, and one for the RTS ratio), to compare the methods to an equal-weighted baseline. The portfolio construction methods tested are minimum variance optimization, risk parity optimization, random weighting and value weighting.

My main finding is that the portfolio construction method can have a statistically significant effect on the skewness of the long-run returns, but this effect is not economically significant for small portfolios (with less than ten stocks), because for such a portfolio the benefit of diversifying better outweighs the benefit of changing the way it is constructed. For larger portfolios (with at least 20 stocks) it can be worth changing from an equal-weighted approach to a minimum variance or risk parity approach. Most importantly, investors should make sure they are not underdiversified before worrying about how they construct their portfolio.

These results are robust when considering different stock market sectors individually, which implies that the minimum variance and risk parity portfolios offer a fundamental improvement that is not a product of random chance. However, I find that a value-weighted portfolio produces returns whose skewness can be both higher and lower than that of the equal-weighted portfolio, depending on the sector.

To end this section, I want to make clear that when I write "skewness", I generally refer to the third *standardized* moment, not to the third central moment. The reason I make this explicit is that I have seen both the standardized and central third moment being referred to as skew in the literature, which can be confusing.

2 Literature Review

Despite the apparent neglect of skewness in the financial literature when compared to the mean and variance, there is still a good amount of highly interesting research on the topic. I aim to give a short overview here. As already mentioned in the Introduction 1, Bessembinder et al. (2023) and Farago and Hjalmarsson (2023a) argue that understanding skewness is important for institutional and individual investors. In short, the reasoning is that only estimating the mean and variance of returns is insufficient to make investment decisions because the majority of outcomes will fall below the mean in the presence of skew, and that this problem is exacerbated over long investment horizons.

Perhaps the most striking result comes from Bessembinder (2018), who shows that long-run US stock return outcomes are heavily skewed. This is reflected in his empirical finding that 4% of stocks explain the net gain of the US stock market over US Treasury bills since 1926, as the remaining 96% collectively matched Treasury bills. A few years later, Bessembinder et al. (2023) find even stronger evidence in the global sample that stock market wealth creation is concentrated in a few top-performing stocks, and that long-run individual stock return outcomes in the global sample are also heavily skewed. These findings suggest that investors should worry about skewness if they care about the long-run outcome of their investment strategy because the distribution of returns is far from symmetric empirically, especially over long investment horizons. As Bessembinder (2018) states himself, he is not the first to uncover the fact that individual stock returns are positively skewed (that fact has been known since at least Simkowitz and Beedles (1978)). He is however one of the first who investigate *long-term* skewness, which is also part of my own research question.

Farago and Hjalmarsson (2019) offer an explanation by showing that multiplicative compounding is what induces the extreme positive skew into individual stock returns. They find that this induced skewness increases with the investment horizon and the standard deviation of short-run returns, and that it can be greatly reduced by diversifying. This is backed up by Bessembinder (2018), who provides simulation evidence that skewness arises through compounding even if the short-run returns are symmetric. Additionally, Farago and Hjalmarsson (2019) state that the probability that a rebalanced portfolio performs better than its best individual constituent in the long-run can be significant, which implies that rebalanced portfolios are less sensitive to missing out on the top-performers.

Farago and Hjalmarsson (2023b) dive deeper into the rebalancing question and go on to show that rebalancing has a great effect on long-run compound portfolio returns. According to them, buy-and-hold portfolios gradually lose diversification as long-run skewness in individual stock returns leads to increasingly concentrated positions. Since periodic rebalancing prevents that gradual loss of diversification, they find that rebalancing greatly reduces skewness. This is consistent with my findings.

Discussion about the higher order moments of return distributions ranges back at least half a century to Arditti (1967), who discusses a Taylor series expansion of investor utility, which makes use of the first three central moments of the investor's return distribution. In a direct criticism of Arditti (1967) however, Levy (1969) states that using only the first three central moments is a problem, because the higher central moments (i.e., fourth and above) cannot be safely ignored in general. These authors later work together to study the relationship between the first three central moments of single period returns and those of multiperiod returns in Arditti and Levy (1975). They find the same relationship as Farago and Hjalmarsson (2019) and Bessembinder (2018) many years later, namely that skewness rises with the investment horizon, and that skewness arises even if the single period returns are symmetric.

To summarize, the literature discussed above suggests that:

- Estimating only the mean and variance of a return distribution is insufficient to make investment decisions.
- Individual stock returns are highly skewed, which is a consequence of multiplicative compounding of returns.
- Investors should diversify to reduce skewness.
- Portfolios should be rebalanced periodically to reduce skewness and improve outcomes.
- Skewness increases with the investment horizon, and is therefore more of a concern for long-term investors.

It seems that whether the method used to construct a stock portfolio can have a significant impact on the skewness of long-run returns seems to be unexplored and relevant to the broader discussion in the literature.

3 Data

My Python code is available here. The relevant file in the context of this section is `data.ipynb`. The code merely executes what is discussed in this section, and it is commented to the best of my ability.

3.1 Source

I use the CRSP/Compustat merged database accessed through the Wharton Research Data Service. This is one of the data sources Bessembinder et al. (2023) also use. In particular, I use daily data from April 1998 to January 2026. A higher sampling frequency increases the accuracy of the covariance estimates, hence why I use daily data. In comparison, Bessembinder (2018) and Bessembinder et al. (2023) use monthly data. No fundamental data was needed.

I use the following features of the aforementioned database: LPERMNO (Permanent Security Number), LPERMCO (Permanent Company Number), GVKEY (Global Company Key), CONM (Company Name), CONML (Company Legal Name), CURCDD (Daily ISO Currency Code), ADRRC (Daily ADR Ratio), CSHOC (Shares Outstanding), CSHTRD (Daily Trading Volume), EXCHG (Stock Exchange Code), TPCI (Issue Type Code), DLRSN (Delisting Reason), GSECTOR (GIC Sector Code), GGROUP (GIC Industry Group Code), GIND (GIC Industry Code), AJEXDI (Cumulative Adjustment Factor by Ex-Date), CSHOC (Shares Outstanding), PRCCD (Daily Close Price), and finally TRFD (Daily Total Return Factor).

3.2 Cleaning

For data cleaning, I generally follow the methodology outlined in the Data Appendix A of Bessembinder et al. (2023), which details the selection of common stocks and numerous other filters. However, I deviate from it where appropriate. Every step is outlined below (in the order the manipulations were applied to the dataset).

1. Cut the dataset at the end of March 1998, because the number of shares outstanding (CSHOC) is not available prior to that date.

2. Selecting common stocks: The issue type code (TPCI) must be either 0 (common stock) or

F (depositary receipt). The inclusion of ADRs implies that non-US stocks are also present in the data set. However, everything is traded in US Dollars, so a currency adjustment is not required.

3. Filtering by exchanges: The stock exchange code (EXCHG) cannot be one of 127 (broker), 283 (fund manager), 157 (fund managers), 229 (OTC), 13 (OTC bulletin-board), 19 (other OTC), 0 (subsidiary/private), 20 (unlisted evaluated equity), 269 (unlisted securities market), 1 (non-traded company or security), or one of 341, 348, 342, 347 (stock connect). After applying this and the previous filters the following stock exchanges remain: 11 (New York Stock Exchange), 14 (Nasdaq Stock Market), 12 (NYSE American), 17 (NYSE Arca), 18 (Nasdaq PHLX), 16 (NYSE Chicago), 21 (Cboe BZX Exchange), and 15 (Nasdaq BX).

4. Exclude any security whose industry code (GIND) is equal to 402040 (Real Estate Investment Trusts). 49 securities are excluded in this step.

5. Exclude funds and trusts, unless they are a bank (GGROUP = 4010). In particular, I remove any security whose company (legal) name (CONM and CONML) include any of the following patterns: "_fund", "_trust", "venture_capital_trust", "_vct", or "_reit". The underscores signify a space. I identify 1267 banks, and exclude 178 funds and trusts.

6. ADR ratio correction: If the security is a depositary receipt, the following transformation is applied: $CSHOC = CSHOC/ADRRC$. The number of shares outstanding (CSHOC) must be corrected with the ADR ratio (ADRRC).

7. Deal with potentially re-used permanent security numbers (LPERMNO). Securities with identical number, but different company key (GVKEY), are considered separate securities. There are 387 re-used security numbers.

8. Pivot the data from long to wide format to facilitate further operations.

9. Drop any stock whose adjustment factor (AJEXDI) ever becomes zero or negative. The reason for this is straight forward. The return calculation requires division by this factor and negative share prices make no sense. At this point there are 14'099 stocks remaining.

Another important thing to note is that by visual inspection, I find no evidence for frequent erroneous decimal shifts or other obvious data errors.

3.3 Calculations

The market capitalization SZ at time t is computed as

$$SZ(t) = PRCCD(t) \cdot CSHOC(t)$$

, the daily return RET (adjusted and with dividends) as

$$RET(t) = \frac{PRCCD(t) \ / \ AJEXDI(t) \cdot TRFD(t)}{PRCCD(t-1) \ / \ AJEXDI(t-1) \cdot TRFD(t-1)} - 1$$

, and the dollar volume DVOL as

$$DVOL(t) = CSHTRD(t) \cdot PRCCD(t).$$

3.4 Further Filters and Data Manipulations

1. Delete the first month of data (first 21 trading days) for each stock. That is, set the returns (RET) of that stock on these dates to N/A. This is done to reduce effects associated with an IPO. Cutting the first month ensures that the stocks enters the dataset only once it is trading under more normal conditions.

2. Missing returns (RET) in the "inside" area (between the first and last non-missing return observations) are filled with zeros. This implicitly assumes that a missing return means that the price did not change. 1.336% of returns are filled.

3. Missing market capitalizations (SZ) are forward filled (in the "inside" area). 1.246% of market capitalizations are filled.

4. Missing sector codes (GSECTOR) are forward filled (in the "inside" area). Forward filling or backward filling is unlikely to make a difference here, since the jump from one sector to another in the data (from one day to the next) is an approximation for the smooth underlying process that the company goes through while changing sectors. 1.252% of codes are filled.

5. Drop all dates where the returns (RET) are zero for every stock in the dataset.

6. The number of zero returns is substantial. I conjecture that this is due to low liquidity/trading volume (without a price update from one day to the next, the return is zero).

It is not due to filled missing values of prior steps, as the percentage of zero returns is far greater than the percentage of values filled. If too many returns are zero, covariance estimates become unreliable. Therefore, I decided to exclude any stocks whose percentage of zero returns exceeds 20%. I arrive at this number through economic intuition. If on average there is no price update on one day per week, then that is acceptable. A total of 43.35% of the stocks are dropped, and 7'987 remain.

7. Delete the remaining history for any stock whose market capitalization (SZ) drops below one million, or whose share price drops below 0.01 USD. This is done because at this point the stock in question is considered to be bankrupt.

8. Drop stocks who now have less than 18 months of data. We require at least twelve months of historical data for the covariance estimation, and at least another three months of returns for the holding period. 7'175 stocks remain after this filter.

9. Drop stocks with unreliable data, i.e. stocks who have a gap of more than one month (21 trading days) without a price update. 7'113 stocks remain after this filter.

10. Filter shares to avoid double counting based on average dollar trading volume (DVOL). If a company has multiple shares associated with it (i.e., when the company number (LPERMCO) is the same for two or more security numbers (LPERMNO)), the security with the highest average dollar trading volume is kept, and the others are dropped. There is an exception to this rule however: If there is no overlap between the securities (indicating that they did not exist simultaneously), they are kept. 98 duplicates are dropped, 7'015 stocks remain.

11. In cases where the delisting reason is either liquidation or bankruptcy (DLRSN $= 2, 3$), the final return (RET) is set to -30% and the market capitalization (SZ) is set to 70% of the value before the delisting.

12. Finally, there are a some outliers that should be dealt with. There are two stocks whose lifetime standard deviation of daily returns is above 50%, and there are eleven stocks whose maximum daily returns exceeded 500%. These stocks are excluded. I conjecture that the reason for these extreme values are errors in the adjustment factor (AJEXDI).

3.5 Examining the Data

It is always a good idea to visualize the data and look at its summary statistics, because that helps to spot obvious errors. Figure 1 shows the distribution of daily returns and Figure 2 shows a scatter plot of each stock's mean and standard deviation of daily returns.

The minimum and maximum daily returns are neither the product of my own manipulations on the data, nor errors in the database. They are extraordinary, yet real events. The involved companies are Point Therapeutics Inc (LPERMNO 80452) on the 10th of October 2007 with a return of 496.33%, and Ocean Rig UDW Inc (LPERMNO 13013) on the 22nd of September 2017 with a return of −96.52%.

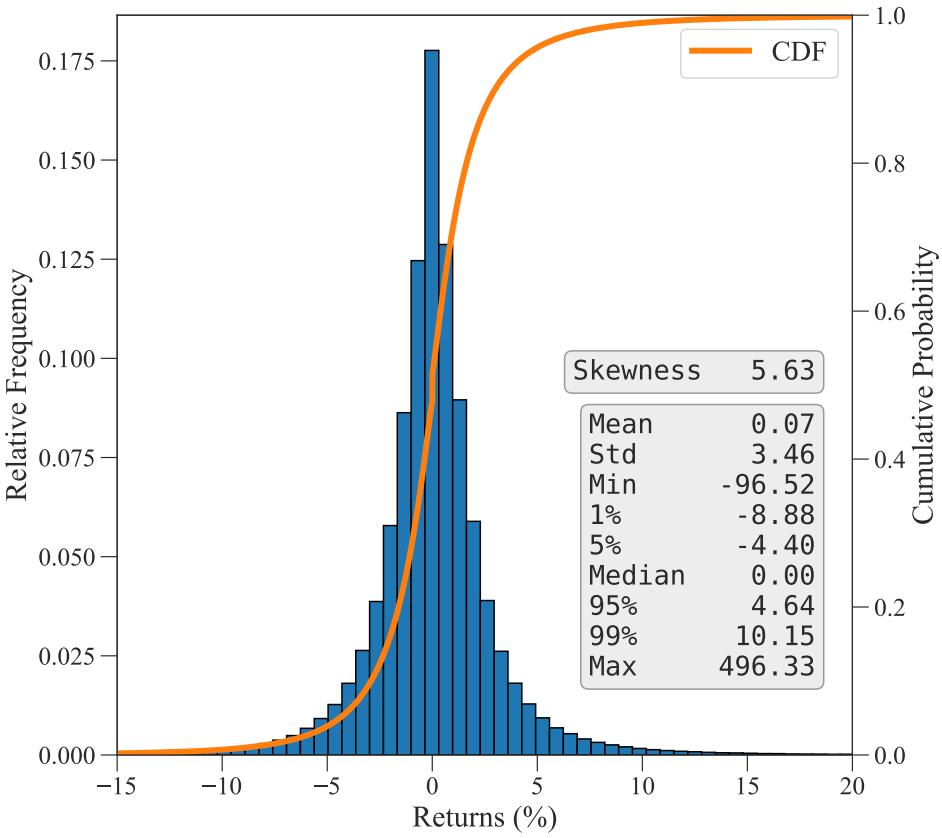

Figure 1: **Distribution of daily returns across all stocks in the cleaned dataset.** The cumulative distribution function is shown in orange and the histogram in blue. The distribution looks as expected, with a mean which is slightly above zero and a median which is exactly zero. The distribution is skewed positively. The standard deviation, along with the minimum return, the maximum return, and 1st, 5th, 95th and 99th percentiles are also given in the grey box. All statistics except skewness are given in %.

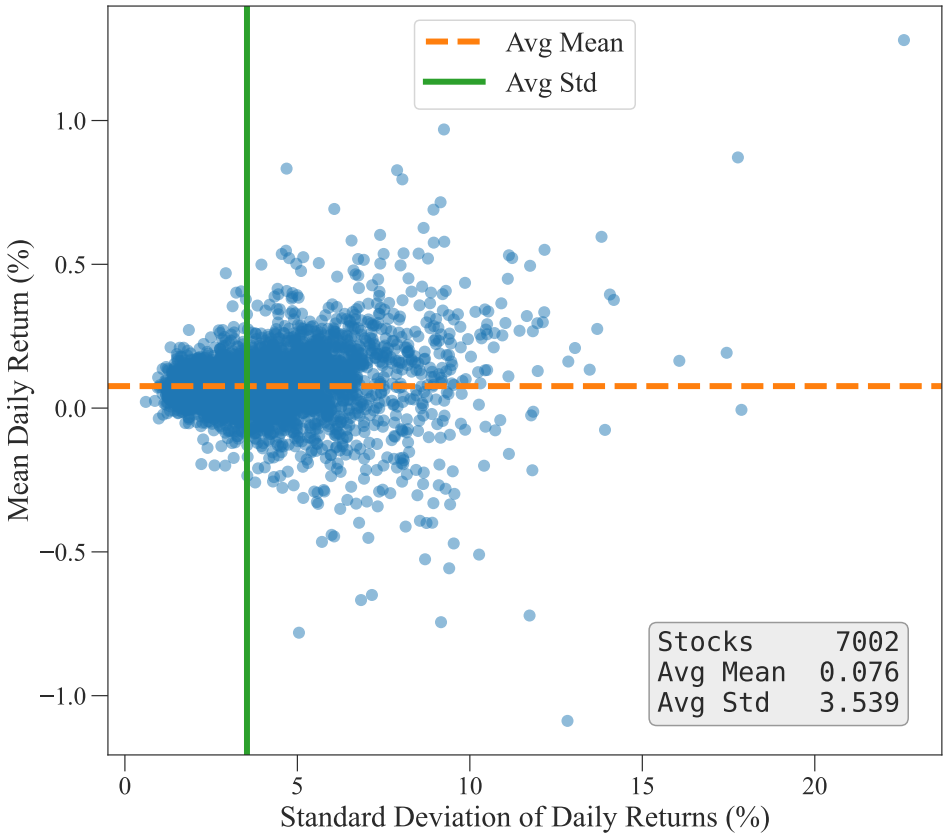

Figure 2: **Scatter plot of each stock's mean and standard deviation of daily returns.** There are 7'002 stocks in the dataset, the average mean of daily returns is 0.076% and the average volatility of daily returns is 3.539%. The point cloud looks as expected with no obvious data anomalies.

4 Methodology

My Python code is available here. The relevant files in the context of this section are `customclass.py` and `regression.ipynb`. The code merely executes what is discussed in this section, and it is commented to the best of my ability.

4.1 Portfolio Construction Methods

What follows is a brief description of the portfolio construction methods I compare against each other.

- *Equal weighting* (Eq). The weight for each stock is given by $1/S$, where S is the size of the portfolio.
- *Value weighting* (Val). The weight of a stock is determined by its relative market capitalization among the stocks in the portfolio.
- *Random weighting* (Rand). The weight of a stock is the product of a random draw from a uniform distribution. The resulting weights are then normalized to ensure that they sum to 1.
- *Minimum variance optimization* (Min Var). The portfolio weights are determined by minimizing the expected variance of the portfolio which is given by $w^T \cdot \Sigma \cdot w$, where w is the vector of portfolio weights and Σ is the covariance matrix of returns. The formula for the portfolio variance has its roots in Markowitz (1952), but I use the matrix notation. This evidently requires estimating the covariance matrix, which I shrink (linearly) according to Ledoit and Wolf (2004). The look-back period for covariance estimation is twelve months of daily returns. Additionally, I consider the long-only case (weights must be positive).

  During experimentation, it became clear that there is a need to constrain the portfolio weights somehow to avoid extreme solutions. It is a well-known fact that minimum variance optimization can lead to unstable solutions. For further reading, see Lopez de Prado (2016) who talks about what he calls Markowitz' Curse: The more correlated the investments, the greater the need for diversification, and yet the more likely there will be unstable solutions. My approach to this problem is to add a regularization term to the

objective function, which is now

$$\arg\min_{w} \quad w^T \cdot \Sigma \cdot w \; + \; L \cdot \sum_{i=1}^{S} \left(w_i - \frac{1}{S} \right)^2$$

, where w_i is the i-th weight and S is the portfolio size. The regularization parameter L controls how much the portfolio weights are allowed to deviate from the equal-weighted case. A value of $L = 0.002$ worked well in practice.

- *Risk parity optimization* (Risk Par). The portfolio weights are determined by solving the equation $\Sigma \cdot w = b/w$, where b is the vector of risk budgets of length S (for risk parity this is a vector filled with entries $1/S$). This problem and an efficient algorithm to solve it are outlined in detail by Spinu (2013). Risk parity optimization also requires estimating the covariance matrix, which I also shrink (linearly) according to Ledoit and Wolf (2004). The look-back period is the same as for the minimum variance optimization, and I also consider the long-only case.

Further things to note are that all portfolios are always invested 100% (there are no cash reserves and there is no leverage), and that the shrunk covariance matrix is computed using the `scikit-learn` implementation of Ledoit and Wolf (2004).

4.2 Distribution Shape

Since the goal is to study the shape of long-run return distributions, I must make clear that "shape" in this context refers to the mean, variance (or standard deviation), and skewness. The third standardized moment (skewness) is of special interest because it seems to be neglected in the literature, which primarily focuses on mean and variance. The `scipy` implementation I use computes the unbiased sample skewness G_1 according to the formula

$$G_1 = \frac{\sqrt{N(N-1)}}{N-2} \cdot \frac{m_3}{m_2^{3/2}}$$

, where N is the sample size and m_i is the i-th central moment of the return distribution. This formula, along with other definitions of sample skewness, can be found in Joanes and Gill (1998).

4.3 Return-To-Shape Ratio

The ratio defined below can be seen as a proxy for investor utility, and can be used to compare the portfolio construction methods in terms of quantities that investors typically care about. A simple approach like this is sufficient for the purposes of my analysis. I define the return-to-shape (RTS) ratio as follows:

$$\mathrm{RTS}\Big(\mu,\, \sigma,\, G_1\Big) = \frac{\mu}{\sigma \cdot [\, 1 + \max(0,\, G_1)\,]}$$

The ratio increases with the mean return μ, but falls with the standard deviation σ and the skewness G_1. Importantly, skewness is penalized less if the standard deviation is low. This makes intuitive sense, as skewness is irrelevant in the absence of risk. Negative skewness is neither penalized nor rewarded. The reason for this is that negative skewness does not cause the same problem that positive skewness causes; if investors base their decisions on the mean of negatively skewed outcome distributions, then most outcomes will actually exceed their expectation. I never observed negative skew throughout my analysis.

The RTS ratio can be viewed as a modified version of the reward-to-variability ratio proposed by Sharpe (1966) (also known as the Sharpe ratio). Since the RTS ratio is computed based on the statistics of a bootstrapped distribution whose outcomes come from different time periods with different interest rate environments, it would not make sense to subtract a benchmark rate (e.g., the risk-free rate) in the numerator of the ratio, since it is not clear what that rate should be. Subtracting the appropriate risk-free rates from the individual portfolio returns during the bootstrap procedure to go from total return to excess returns does not make sense either, since compounding the excess returns instead of the total return leads to outcomes that no longer reflect the actual return investors receive. Because of this, I decided to make the simplifying assumption that the risk-free (benchmark) rate is zero.

There are previous (and more sophisticated) attempts to extend or adjust the Sharpe ratio with a measure of skewness. As an example, see Zakamouline and Koekebakker (2009) who derive such metrics. For the purposes of my analysis, a simple approach with an easy interpretation is sufficient.

Finally, the RTS ratio is specific to the time horizon. For a comparison between two return distributions to be valid, they must have the same underlying investment horizon, because G_1, μ, and σ depend on it. Since the regression models control for the horizon, this is not a problem.

4.4 Bootstrapping

I employ the bootstrapping method used by Bessembinder (2018) for equal, value, and random weight portfolios. For minimum variance and risk parity portfolios I modify the procedure slightly, because they require a look-back period to estimate the covariance matrix.

Let S be the portfolio size and let H be the investment horizon in years. For each 3-month interval in the dataset, a set of S stocks is randomly selected. The weights are then computed according to some method (equal-weighting, minimum variance etc.) and its three-month return computed by compounding the daily portfolio returns until the beginning of the next three-month period. One bootstrapped long-run outcome is equal to the cumulative return over H years obtained in this way. By repeating this process many times, a distribution of long-run outcomes is obtained (with the same size, horizon and method!). For an example of such a distribution, see Figure 3. The skewness and the RTS ratio of the bootstrapped distributions form the dependent variables for their respective regression models.

This procedure implies a rebalancing/re-optimization interval of three months. The reason for choosing a three- rather than one-month interval are computational constraints. Most of the computing time is spent optimizing the portfolios for the minimum variance and risk parity methods. Re-optimizing every third month instead of every month decreases the number of optimizations considerably, which allows for more portfolios to be bootstrapped, which increases the accuracy of the skewness estimates. Farago and Hjalmarsson (2023b) show that rebalancing per se (and the choice of the interval) fundamentally changes the distribution of long-run outcomes. This is why I use the same interval for all methods.

For investment horizons H that are shorter than the number of years in the dataset, the starting month is varied to cover the entire dataset.

The seemingly high cumulative returns of the risk parity portfolios (on the order of a 30-fold increase over 25 years) in Figure 3 are in line with what Farago and Hjalmarsson (2023b) find for their periodically rebalanced equal-weighted portfolios (they find a 20-fold increase over 25 years). It is important to note that the time periods of the underlying historical returns do not match between mine and their study.

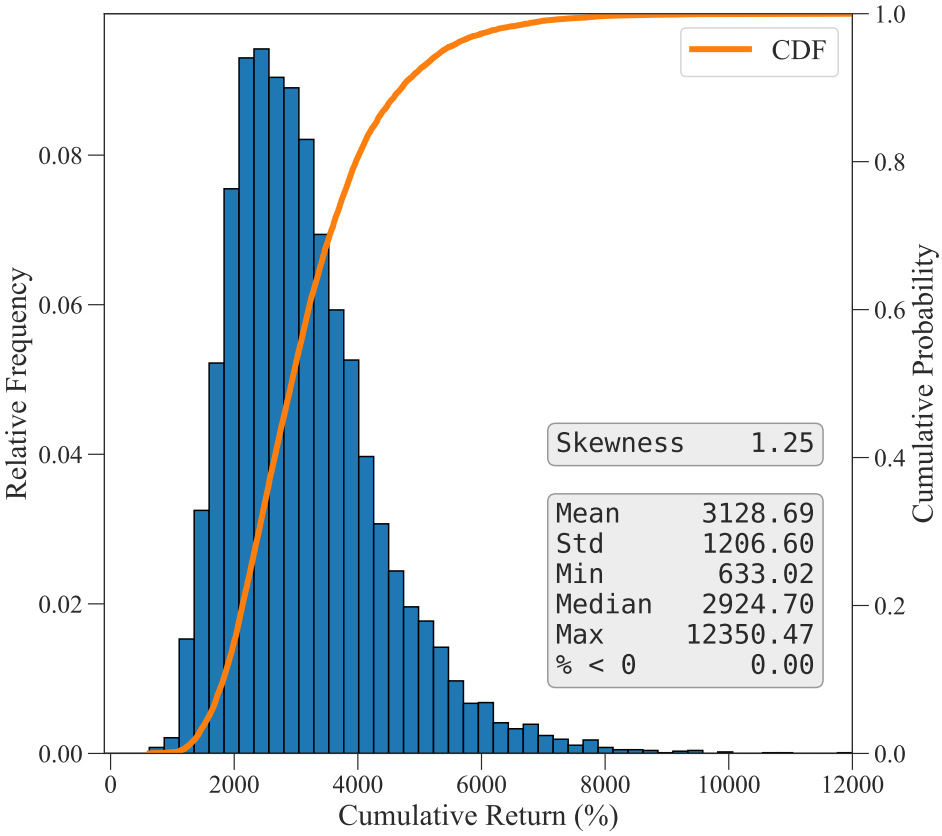

Figure 3: **Distribution of cumulative returns (i.e., long-run outcomes) of 10'000 bootstrapped 32-stock risk parity portfolios with an investment horizon of 25 years.** The cumulative distribution function is shown in orange and the histogram in blue. The distribution is positively skewed. The mean, standard deviation, minimum return, maximum return, median, as well as the percentage of returns below zero are shown in the grey box. All values except skewness are given in %.

4.5 Regression Model

To answer my research question, I run two regressions, one for skewness G_1 and one for the RTS ratio. It is appropriate to use two different models, because the portfolio size affects the dependent variables differently. I find that skewness is best described using the inverse of the portfolio size, while for the RTS ratio a combination of linear and quadratic size terms fit best. The models to be estimated are

$$G_1 \sim \beta_0 + \beta_1 \cdot \frac{1}{S} + \beta_2 \cdot H + \beta_3 \cdot \text{Val} + \beta_4 \cdot \text{Min Var} + \beta_5 \cdot \text{Risk Par} + \beta_6 \cdot \text{Rand}$$

$$\text{RTS} \sim \beta_0 + \beta_1 \cdot S + \beta_2 \cdot S^2 + \beta_3 \cdot H + \beta_4 \cdot \text{Val} + \beta_5 \cdot \text{Min Var} + \beta_6 \cdot \text{Risk Par} + \beta_7 \cdot \text{Rand}$$

, where S and H stand for the portfolio size and investment horizon. The remaining variables are dummies and refer to the portfolio construction methods (equal weighting (Eq), value weight-

ing (Val), minimum variance optimization (Min Var), risk parity optimization (Risk Par), and random weighting (Rand)). The equal-weighted case is omitted in the model to avoid perfect multicollinearity. Eq therefore represents the baseline against which the other methods are measured. The random weighting scheme is included to serve as a sanity check; it should produce higher long-run skewness than the equal-weighted baseline because it will ceteris paribus produce more concentrated portfolios with higher volatility.

The sign of the coefficients of the dummy variables in the regression on skewness are of particular interest. The sign primarily depends on two factors. First, whether the standard deviation of short-run portfolio returns is lower or higher than that of the equal-weighted baseline, and second, the degree of concentration when compared to the equal-weighted baseline. There is a trade-off between reduced volatility (which decreases skew), and higher concentration (which increases skew). These two facts are documented by Farago and Hjalmarsson (2019). The minimum variance optimization for example will produce a portfolio that is more concentrated than the equal-weighted baseline, but with a lower volatility. Which effect is dominant is not obvious.

A dataset for a regression analysis is constructed by running the bootstrap procedure for a variety of different portfolio sizes (S), horizons (H) and methods (Eq, Val, Min Var, Risk Par, Rand), and then computing their respective skewness, mean and standard deviation. Refer to Table 1 for an illustration. Each row corresponds to one bootstrapped distribution of outcomes with its own unique combination of S, H and portfolio construction method. The sizes and horizons for which the bootstrap procedure is conducted are:

$$S \in \{2,\ 5,\ 8,\ 11,\ 14,\ 17,\ 20,\ 23,\ 26,\ 29,\ 32\}$$

$$H \in \{1,\ 4,\ 7,\ 10,\ 13,\ 16,\ 19,\ 22,\ 25\}$$

Since G_1, μ, and σ depend on the investment horizon, it is important to control for the horizon in the regression model. This is the main reason why it is included. Its inclusion also implies that these statistics should not be transformed to monthly or daily values.

Each row in Table 1 is the result of 2'000 bootstrapped portfolios. In comparison, Bessembinder (2018) bootstraps 20'000 portfolios. Since my analysis demands much more computing power however, that number is not achievable for me. The skew estimate improves with the number of samples however, so I set the number to be as high as possible.

G_1	μ	σ	S	H	Eq	Val	Min Var	Risk Par	Rand
2.74	534.5	530.9	5	13	1	0	0	0	0
1.41	271.9	140.6	14	10	0	0	0	1	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Table 1: **The data table produced by the bootstrap procedure.** Two rows of the actual data were randomly selected to illustrate. The first row shows the skewness G_1, the mean return μ (in %), and the standard deviation σ (in %) of the return distribution of equal-weighted (Eq) portfolios with $S = 5$ stocks over a horizon of $H = 13$ years. The whole table has 495 rows, which is equal to the number of methods (5) times the number of different sizes (11) times the number of different horizons (9), and each row is the result of 2'000 bootstrapped portfolios.

At this point it is appropriate to mention an implicit assumption I made when specifying the regression models, which is that the coefficients for $1/S$, S, S^2 and H in their respective models are the same for all portfolio construction methods. I do not include interaction terms because it simplifies the model, and because their inclusion made almost no difference when exploring different model setups.

Finally, the independent variables $1/S$, S, S^2 and H were centered for the regression. This is important for the interpretation of the results, as the coefficient of the constant becomes the predicted value of the dependent variable when these predictors are at their mean values, rather than when they are all equal to zero.

4.6 Robustness Tests

Robustness tests are conducted to check the validity of the regression results. I run the entire bootstrapping procedure and the same regression for different subsets of the dataset, which each correspond to one sector. For example, one of the tests is run on stocks from the energy sector (GSECTOR 10), and another on utilities (GSECTOR 55). Each sector is tested separately in this way.

The goal is to see if the coefficient estimates are stable in the sense that they have the same magnitude and sign when filtered by sectors, as in the unfiltered dataset.

5 Results and Discussion

The bootstrapped data tables are available here. I invite the reader to download them and to run their own regressions. This hopefully reduces the barrier for further analysis, as this eliminates the need for the CRSP/Compustat dataset (which I am not allowed to make publicly available) and also eliminates the need to run the bootstrapping procedure again (which took multiple days on a decently powerful desktop computer). To give an idea where to start, one might calculate the Sharpe ratios (instead of the RTS ratio) and use that as the dependent variable.

5.1 Visualization

Before we turn our attention to the coefficients and tables in the next subsections, I want to begin with a visual analysis of the results. The three relevant figures are Figure 4, Figure 5 and Figure 6. Their captions contain the most important information.

Figure 4 is obtained by plotting skewness G_1, portfolio size S and investment horizon H of each row in the regression data table (see Table 1) where the dummy variable Eq is equal to 1, against each other. So, each point represents one bootstrapped distribution of outcomes, much like the one shown in Figure 3. Beyond what is stated in the caption of Figure 4, I notice that the investment horizon seems to matter more in the context of skew for small portfolios. It seems therefore that diversification reduces the sensitivity to the skew-inducing effect of compounding returns that Farago and Hjalmarsson (2019) mention, possibly because a diversified portfolio's single period returns are less volatile, which reduces skewness according to that same paper.

Figure 5 is obtained by first plotting skewness G_1 and portfolio size S of each row in the regression data table (see Table 1) against each other, which results in the grey dots. Then, the lines are constructed using the coefficients from Table 2. It is important to note that this is at average horizon, since H was controlled for and centered. A quick sanity check reveals that the model holds up well in comparison to Bessembinder (2018), who reports that the returns of bootstrapped equal-weighted 25-stock portfolios over a horizon of 10-years (which is close to the average horizon of 13 years in my analysis) is around 1.64. Given that the blue line (Eq) predicts a skewness of (visually) approximately 1.60 around a portfolio size of 25, the plot makes sense. I argue that while there is a statistically significant difference in skewness between

the portfolio construction methods, it is easier to simply add more stocks to a portfolio to reduce skewness. However, the caveat is that since the marginal decrease in skew falls with each additional stock, there inevitably comes a point at which an investor might consider a different portfolio construction method to decrease skew further.

Figure 6 is obtained analogously as Figure 5, but here the coefficients are pulled from Table 3. For the interpretation it is important to know that a higher RTS ratio is better, and that the lines are also shown at the average investment horizon. The plot shows that for small portfolios the construction method is less important than diversifying more. However, for larger portfolios, the marginal increase in the RTS ratio for each additional stock seems to vanish, at which point the method could be the only way to increase it. This needs to be explored further however, by examining portfolios that are larger than the ones tested here, to avoid making an extrapolation error. The reason for stopping at a portfolio size of 32 are computational constraints. I expand on this in Subsection 5.4.

To summarize, we see the same pattern for both Figure 5 and 6, which is that investors should first think about diversification, and only then about how to construct their portfolio.

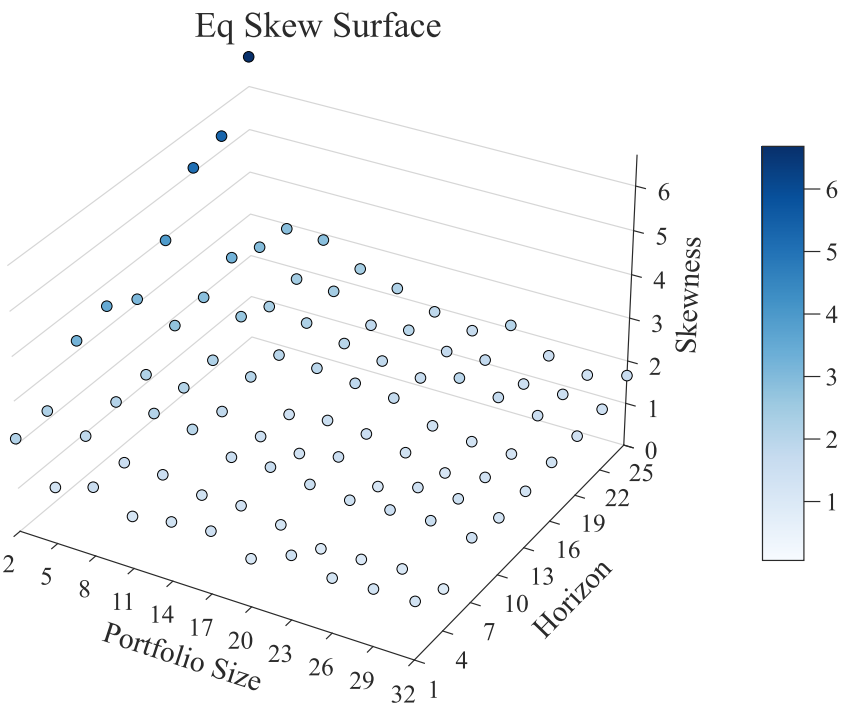

Figure 4: **The skew surface for equal-weighted portfolios.** Each point shows the skewness of the return distribution of 2'000 bootstrapped equal-weighted (Eq) portfolios with a specific size and a specific investment horizon (in years). The skewness rises with the investment horizon and falls with the number of stocks in the portfolio. This means that skewness can be reduced by diversifying. The skew surfaces of other portfolio construction methods can be found in the Appendix: Figure 7 (Min Var), Figure 8 (Risk Par), Figure 9 (Val), Figure 10 (Rand). The pattern found here is also present for the other methods.

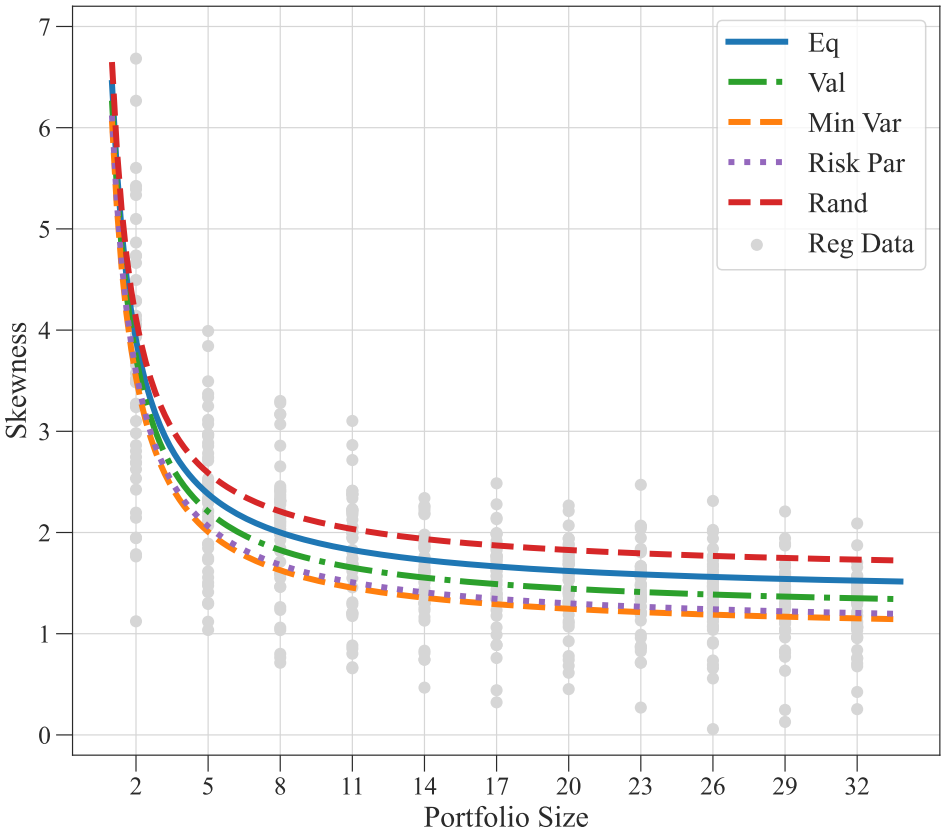

Figure 5: **The relationship between skewness and portfolio size for different portfolio construction methods.** The methods are equal-weighted (Eq), value-weighted (Val), minimum variance optimization (Min Var), risk parity optimization (Risk Par), and random-weighted (Rand). The lines are constructed using the regression coefficients of Table 2. While the portfolio construction method has a statistically significant effect on skewness, it can be argued that these effects are not economically significant for the methods that were tested if the portfolio size is small. Investors should worry more about adding more stocks to their portfolio than about how to construct their portfolio, when thinking about skewness specifically. The data points from the regression data table are drawn in light grey (Reg Data). Skewness is described well by the inverse of the portfolio size.

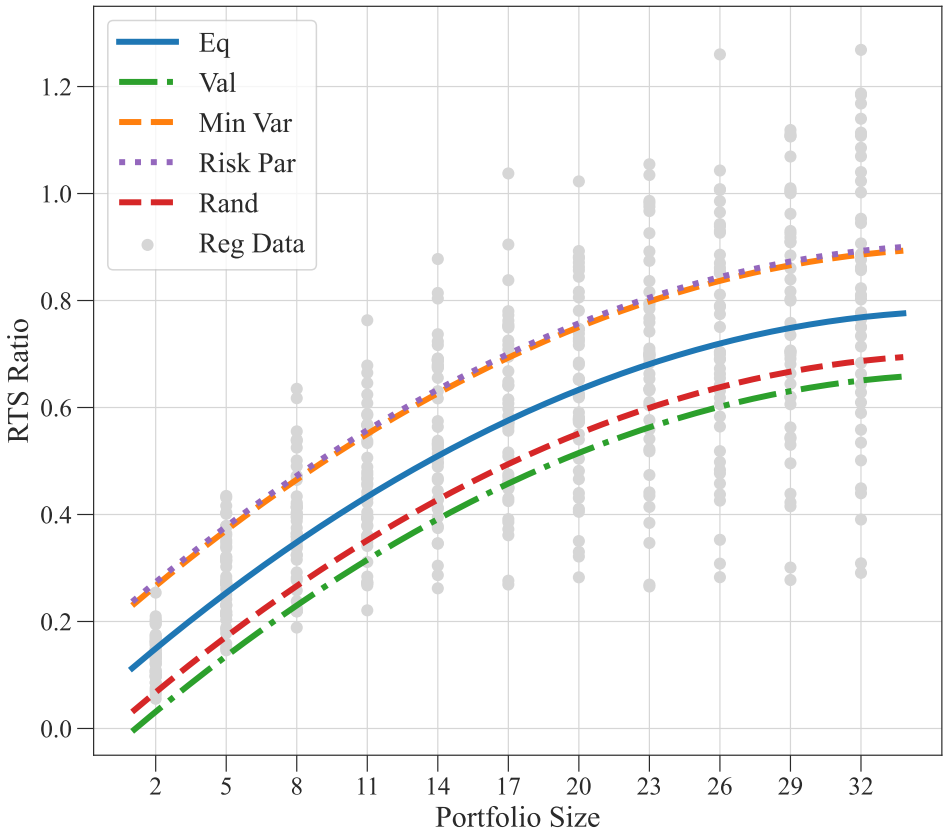

Figure 6: **The relationship between the RTS ratio and portfolio size for different portfolio construction methods.** The methods are equal-weighted (Eq), value-weighted (Val), minimum variance optimization (Min Var), risk parity optimization (Risk Par), and random-weighted (Rand). The lines are constructed using the regression coefficients of Table 3. The portfolio construction method can make a difference, especially when comparing Val with Risk Par, but investors should still diversify first before optimizing. The data points from the regression data table are drawn in light grey (Reg Data).

5.2 Regression Tables

In this subsection I go over the numerical results and how to interpret them. The captions of Tables 2, 3, 4 and 5 hold the most important information. I am not interested in the exact values of the coefficients, but in their sign, relative magnitude, and whether they are statistically and economically significant. The answer is that they are statistically significant, and are economically significant only for large portfolios (when the benefit of further diversification is small).

In short, both Tables 2 and 3 indicate that the portfolio construction method can have a statistically significant impact on the skewness and the RTS ratio of the return distributions. Additionally, they indicate that minimum variance and risk parity portfolios beat the equal-weighted benchmark in both cases, while random-weighting always performs worse than the benchmark (as expected). The value-weighted case is more nuanced. While the returns of value portfolios are less skewed, the RTS ratio is lower, because that reduction in skew seems to come at a considerable cost for this method.

The constant in Table 2 is to be interpreted as the skewness of an equal-weighted portfolio at average investment horizon, when $1/S$ is at its average value. The constant in Table 3 is to be interpreted as the RTS ratio of an equal-weighted portfolio at average investment horizon, when S and S^2 are at their average values.

The interpretation of the method coefficients is that if an investor (ceteris paribus) changes from an equal-weighted portfolio to a minimum variance portfolio for example, the dependent variable (skewness or the RTS ratio respectively) will change by that amount. So, for skewness a negative coefficient is considered good, while for the RTS ratio a positive coefficient is considered good.

I found that the inverse of the portfolio size describes the return skewness well. It is important to keep in mind that just because the coefficient is highly significant does not mean that this is the underlying "true" relationship, but the model fits the data well, as seen in Figure 5.

Skewness Regression Results				
Variable	Coefficient	Standard Error	t-value	p-value
Constant	1.9235***	0.0396	48.6350	0.0000
$1/S$ (centered)	5.0755***	0.1346	37.6985	0.0000
H (centered)	0.0543***	0.0023	23.7938	0.0000
Val	-0.1743**	0.0559	-3.1162	0.0019
Min Var	-0.3730***	0.0559	-6.6693	0.0000
Risk Par	-0.3199***	0.0559	-5.7190	0.0000
Rand	0.2074***	0.0559	3.7089	0.0002

Table 2: **Results for the regression on skewness.** Skewness falls significantly with the inverse of the portfolio size. This implies that adding more stocks to a portfolio is more important for small portfolios. Skewness increases with each additional year of investment. The return distribution of value-weighted, minimum variance and risk parity portfolios are less skewed than that of equal-weighted portfolios. While statistically significant, I argue that this reduction in skewness is not economically significant, however. This is certainly true for small portfolio where diversifying offers the easiest way to decrease skew. All values were rounded to four decimal places. (Significance codes: *** for $p < 0.001$, ** for $p < 0.01$, * for $p < 0.05$, . for $p < 0.1$)

RTS Ratio Regression Results				
Variable	Coefficient	Standard Error	t-value	p-value
Constant	0.5370***	0.0134	40.1882	0.0000
S (centered)	0.0383***	0.0026	14.5569	0.0000
S^2 (centered)	-0.0005***	0.0001	-6.8987	0.0000
H (centered)	0.0039***	0.0008	5.0819	0.0000
Val	-0.1181***	0.0189	-6.2517	0.0000
Min Var	0.1172***	0.0189	6.2013	0.0000
Risk Par	0.1242***	0.0189	6.5730	0.0000
Rand	-0.0817***	0.0189	-4.3242	0.0000

Table 3: **Results for the regression on the RTS ratio.** Each additional stock in the portfolio increases the ratio, but the effect decreases with diversification. This implies that adding more stocks to a portfolio is more important for small portfolios. The ratio also increases with each additional year of investment. Minimum variance and risk parity portfolios achieve a higher ratio than the equal-weighted portfolio, while the value-weighted and random-weighted portfolio fail to match the benchmark. All values were rounded to four decimal places. (Significance codes: *** for $p < 0.001$, ** for $p < 0.01$, * for $p < 0.05$, . for $p < 0.1$)

5.3 Robustness Tests

In the context of the robustness tests I primarily look for whether the sector-specific coefficients have the same sign and magnitude as the main regression coefficients. If that is given, then that bolsters the confidence we can have in the main results. Significance is of less importance here, because when running multiple tests at the same time we expect some of them to be more or less statistically significant by chance.

When looking at Table 4, three facts stand out. First and most importantly, the coefficients for the minimum variance, risk parity, and random portfolios have the same sign and magnitude for all sectors. Second, the coefficients for the value portfolio mostly have the same sign, except for sector 40 (Financials) who stands out with positive and significant coefficient, and the magnitude and significance of the coefficients also varies greatly in the value column. For some reason, the financial sector is special in the context of skew when looked at from a value perspective. Third, the constant, while always significant and positive, has a large range. There is more skew in some sectors, and less in others.

Another three things stand out when looking at Table 5. First and most importantly, the coefficients for the minimum variance, risk parity, and random portfolios have the same sign and magnitude for all sectors. Second, the sign of the coefficients for the value portfolio varies across the column. Finally, the constant, while always significant and positive, has a relatively large range (just like in Table 4).

In general, the robustness test strengthens my confidence in the findings of the previous subsections, at least for the minimum variance, risk parity and random portfolios. The value portfolio is elusive, as the coefficients are not robust across sectors. A straight-forward answer is that the value-weighted case might simply be sensitive to the sector.

Information about where to find the full regression results for each sector can be found in Appendix A.2.

Skewness Regression Sector Comparison					
Sector	Constant	Val	Min Var	Risk Par	Rand
All	1.9235***	-0.1743**	-0.3730***	-0.3199***	0.2074***
10	1.7672***	-0.1737**	-0.1543**	-0.1213*	0.2077***
15	1.7729***	-0.0359	-0.1826**	-0.1388*	0.1903**
20	1.8322***	-0.4608***	-0.3135***	-0.2670***	0.2291***
25	2.0191***	-0.1249.	-0.2789***	-0.2054**	0.2275***
30	1.4372***	-0.3150***	-0.1701***	-0.1828***	0.1913***
35	2.5712***	-0.8899***	-0.4550***	-0.3536***	0.2931***
40	1.2112***	0.3933***	-0.1361**	-0.1377**	0.1030*
45	2.1001***	-0.1440.	-0.1566*	-0.1883*	0.2010*
50	1.8213***	-0.2244***	-0.3923***	-0.2614***	0.1883**
55	0.8033***	0.0417	-0.1526***	-0.1647***	0.1185**
60	1.2294***	-0.0476	-0.1757***	-0.1638***	0.1428**
High	35	40	40	10	35
Low	55	35	35	35	40
Diff	1.7678	1.2831	0.3189	0.2323	0.1901

Table 4: **Skewness Regression Coefficients by Sector.** The methods are value-weighted (Val), minimum variance optimization (Min Var), risk parity optimization (Risk Par), and random-weighted (Rand). They are measured against the equal-weighted baseline. The GIC sector codes are 10 Energy, 15 Materials, 20 Industrials, 25 Consumer Discretionary, 30 Consumer Staples, 35 Health Care, 40 Financials, 45 Information Technology, 50 Communication Services, 55 Utilities, and 60 Real Estate. The coefficients of the "All" Sector are those of Table 2. At the bottom of the table are the sector codes with the highest (High) and lowest (Low) values for each column. The absolute difference (Diff) between highest and lowest values per column are also shown. (Significance codes: *** for $p < 0.001$, ** for $p < 0.01$, * for $p < 0.05$, . for $p < 0.1$)

RTS Ratio Regression Sector Comparison					
Sector	Constant	Val	Min Var	Risk Par	Rand
All	0.5370***	-0.1181***	0.1172***	0.1242***	-0.0817***
10	0.4315***	-0.0274.	0.0264.	0.0466**	-0.0542***
15	0.5649***	-0.1221***	0.0585**	0.0785***	-0.0896***
20	0.5842***	0.0578**	0.0996***	0.1050***	-0.0988***
25	0.4847***	-0.0952***	0.0221	0.0422*	-0.0755***
30	0.7815***	0.1509***	0.1185***	0.1381***	-0.1231***
35	0.3714***	0.1692***	0.1116***	0.0895***	-0.0584***
40	0.7598***	-0.3804***	0.0431.	0.0739**	-0.0860***
45	0.4759***	-0.1550***	0.0312	0.0416.	-0.0688**
50	0.4886***	-0.0490*	0.1111***	0.1075***	-0.0738***
55	1.5937***	-0.3895***	0.2661***	0.2698***	-0.2524***
60	0.8295***	-0.0491.	0.0106	0.0547.	-0.1175***
High	55	35	55	55	10
Low	35	55	60	45	55
Diff	1.2223	0.5586	0.2555	0.2282	0.1981

Table 5: **RTS Ratio Regression Coefficients by Sector.** The other methods are value-weighted (Val), minimum variance optimization (Min Var), risk parity optimization (Risk Par), and random-weighted (Rand). They are measured against the equal-weighted baseline. The GIC sector codes are 10 Energy, 15 Materials, 20 Industrials, 25 Consumer Discretionary, 30 Consumer Staples, 35 Health Care, 40 Financials, 45 Information Technology, 50 Communication Services, 55 Utilities, and 60 Real Estate. The coefficients of the "All" Sector are those of Table 3. At the bottom of the table are the sector codes with the highest (High) and lowest (Low) values for each column. The absolute difference (Diff) between highest and lowest values per column are also shown. (Significance codes: *** for $p < 0.001$, ** for $p < 0.01$, * for $p < 0.05$, . for $p < 0.1$)

5.4 Limitations

There are various limitations I want to make the reader aware of.

- There are some limitations in connection with limited computational resources:
  - The maximum portfolio size was $S = 32$, because the covariance matrix estimation and the algorithms for computing the minimum variance and risk parity portfolios become much slower for larger portfolios when compared to the equal-weighted base case.
  - Skewness, mean and standard deviation estimates are based on distributions of N =2'000 bootstrapped portfolios. Since the estimates improve with the number of samples, a higher N is desirable, but is not feasible because of computational constraints.
- Skewness is difficult to estimate and can be very noisy in my experience (which is visible in the skew surface plots in Figures 4, 7, 8, 9, 10). This is a problem also mentioned by Farago and Hjalmarsson (2023a). I found that trimming any observations more than eight standard deviations from the mean reduced this problem substantially. This makes sense because skewness (G_1) is sensitive to outliers, but it may also introduce a bias. There is a trade-off here. This procedure was done consistently for all skew estimates, not selectively.
- There are a variety of different portfolio construction methods that could also be explored, but whose inclusion would have exceeded the scope of this work. Examples are inverse volatility, inverse market capitalization, momentum, low-beta, high-beta, or various other factor portfolios.
- The look-back period to estimate the covariance matrix is set to one year. There are arguments in favor of a longer window. A more thorough analysis would vary this parameter also to see how it affects things.
- All portfolios are rebalanced/re-optimized every three months. There are arguments for shorter or longer intervals. A more thorough analysis would vary this parameter also to see how it affects things.
- Many stocks are excluded from the dataset because of the need for sufficient historical data for the covariance estimation. Without this filter the dataset would be approximately 10% larger.

- Zero-returns caused by the necessary filling of missing returns impact the covariance estimations and therefore the optimization results of the minimum variance and risk parity portfolios.
- Model-risk: Despite thorough exploration of possible regression models to find those that are both simple and fit the data best, there might be models that explain the dependent variables even better.
- This analysis is restricted to US stock markets.

6 Conclusion

In conclusion, the portfolio construction method can have a statistically significant effect on the skewness of the long-run return distribution, but this effect is not economically significant for small portfolios (with less than ten stocks). For larger portfolios (with at least 20 stocks) it can be worth changing from an equal-weighted approach to a minimum variance or risk parity approach. But further analysis of the behavior of these optimization methods for portfolios with more than 32 stocks (the largest size tested here) would be beneficial to avoid making an extrapolation error. In general, we learned that investors should make sure they are not underdiversified before worrying about how they construct their portfolio.

Sector tests show that these results are robust for the minimum variance, risk parity and random portfolios, while the value-weighted portfolio's effect on skewness and the RTS ratio seems to be sector dependent.

When compared to an equal-weighted baseline the minimum variance and risk parity portfolios produce less skew, while a random-weighted portfolio produces more skew. The value-weighted portfolio's effect on skewness can be positive or negative compared to the baseline, depending on the stock sector. Additionally, the skewness observed in bootstrapped stock portfolios varies between sectors. Some sectors produce outcomes that are more skewed than others.

When comparing the portfolio construction methods against an equal-weighted baseline by using the RTS ratio, which can be seen as a modified Sharpe ratio, I find that the minimum variance and risk parity portfolios offer an improvement, while the random-weighted portfolio performs worse. The value-weighted portfolio's effect here too depends on the sector.

I confirmed the findings of Bessembinder (2018) and Farago and Hjalmarsson (2019), namely that skewness increases with the investment horizon and that it can be reduced by diversifying. Furthermore, I show that diversification is not the only way to decrease skewness, and that the portfolio construction method can matter too.

Further research might focus on the difference between sectors when it comes to skewness, and why the value-weighted approach produces more or less skew than the equal-weighted approach depending on the sector. Another research opportunity is to address the limitations of my analysis. For example, one could examine a larger and more diverse set of portfolio construction

methods with the goal of finding the one that minimizes skewness. There is definitely much to be explored still.

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A Appendix

A.1 More Skew Surfaces

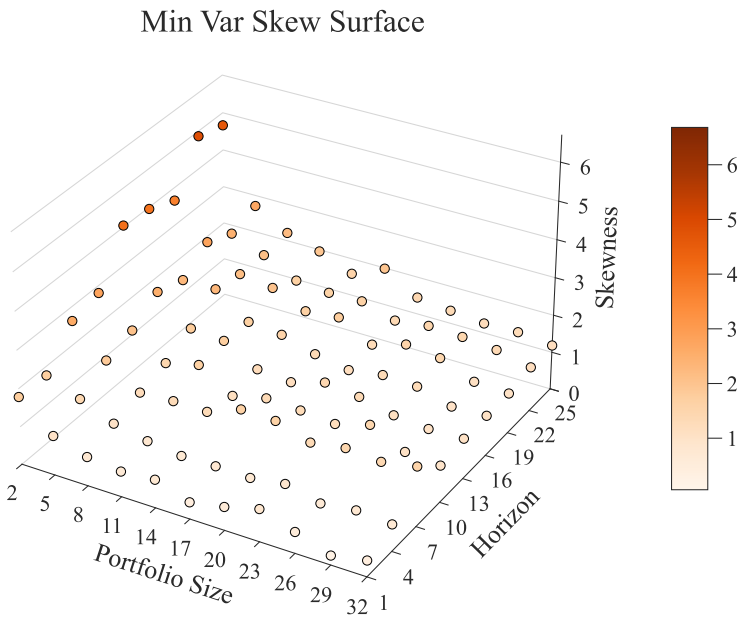

Figure 7: **The skew surface for minimum variance portfolios.** Each point shows the skewness of the return distribution of 2'000 bootstrapped minimum variance (Min Var) portfolios with a specific size and a specific investment horizon (in years). The skewness rises with the investment horizon and falls with the number of stocks in the portfolio.

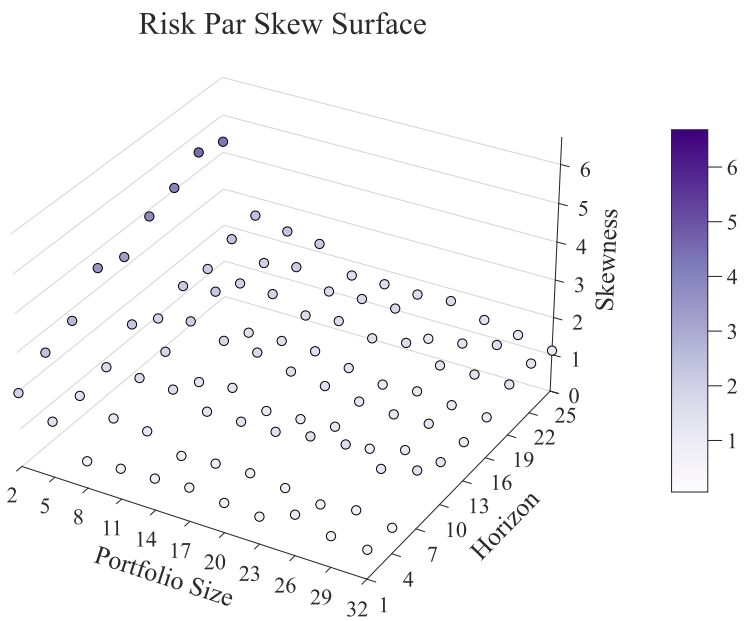

Figure 8: **The skew surface for risk parity portfolios.** Each point shows the skewness of the return distribution of 2'000 bootstrapped risk parity (Risk Par) portfolios with a specific size and a specific investment horizon (in years). The skewness rises with the investment horizon and falls with the number of stocks in the portfolio.

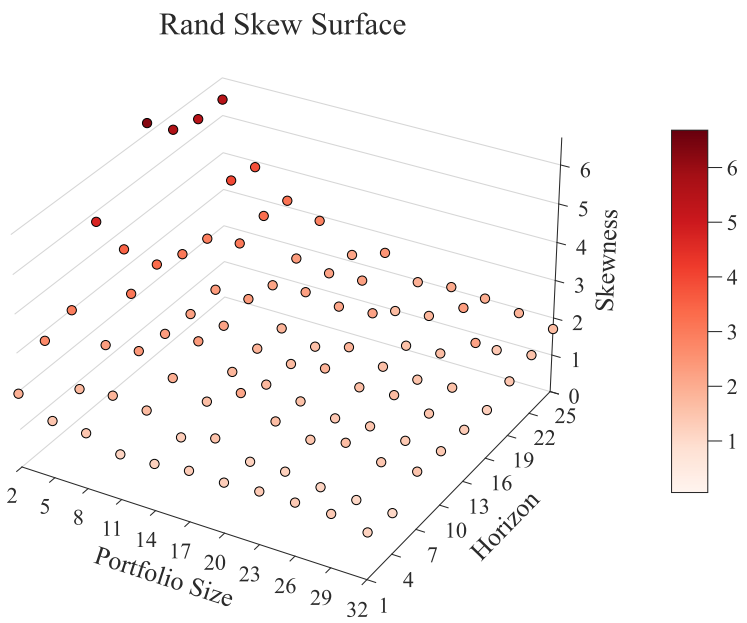

Figure 10: **The skew surface for random-weighted portfolios.** Each point shows the skewness of the return distribution of 2'000 bootstrapped random-weighted (Rand) portfolios with a specific size and a specific investment horizon (in years). The skewness rises with the investment horizon and falls with the number of stocks in the portfolio.

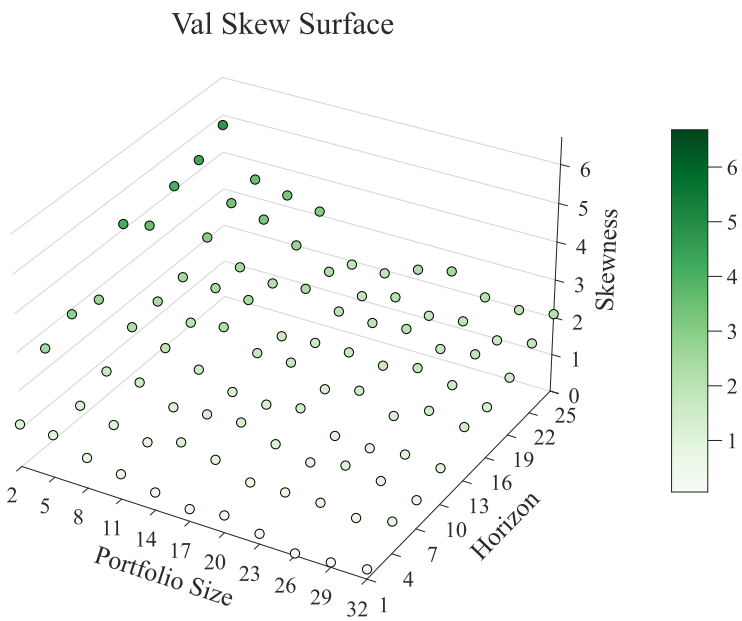

Figure 9: **The skew surface for value-weighted portfolios.** Each point shows the skewness of the return distribution of 2'000 bootstrapped value-weighted (Val) portfolios with a specific size and a specific investment horizon (in years). The skewness rises with the investment horizon and falls with the number of stocks in the portfolio.

A.2 More Regression Tables

The full regression results are available as `.csv` files here. The naming scheme is as follows, the files are named `results_sectorGSECTOR_Y.csv`, where `GSECTOR` is the GIC sector code (e.g., 20 for industrials) and `Y` is the dependent variable of the regression model (e.g., skew). The only exception is sector 0, which denotes the entire dataset (not filtered by sectors).